

- 17** The three main factors that affect the rate of diffusion across a membrane can be expressed by the relationship shown.

rate of diffusion is proportional to $\frac{\text{surface area} \times \text{concentration difference}}{\text{thickness of membrane}}$

Which changes in the factors would result in the rate of diffusion doubling?

- 1 Surface area has doubled.
- 2 Concentration difference has halved.
- 3 Thickness of membrane has doubled.
- 4 Thickness of membrane has halved.

A 1, 2 and 4 **B** 1 and 3 **C** 1 and 4 only **D** 2 and 3